

RAJALAKSHMI ENGINEERING COLLEGE

TEAM NAME	**ANNADATHA AI**
TEAM MEMBERS	NITIN R **241701041** RASHETHA THASNEEM M **241701046** KORAPATI HEMANTH **241701501**

Rajalakshmi Engineering College

Department of Computer Science and Design

GE23627 DESIGN THINKING AND INNOVATION

2026-2027

S.C.O.P.E.S – Team Scope Sheet

Scoping of design challenge When you want to work on real project using Design Thinking, begin the process with a specific and purposeful situation or problem to solve called Design Challenge. The Design Challenge should be doable, clear, actionable and hopefully inspiring. Your problem statement should be an outcome and not a solution.

You can use SCOPES as a tool to brainstorm and frame your problem.

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DOMAIN NAME	**Artificial Intelligence (AI),Machine learning(ML) Computer Vision, Augmented/Mixed Reality (AR/MR), and Smart Agriculture**
DATE	
FACULTY INCHARGE	**DR.P.REVATHY** **Computer Science and Design**

S – Situation / Problem • What is the real situation you want to address? • Where does this problem occur? • Describe without mentioning any solution.	Farmers often face difficulties in identifying crop diseases at an early stage. Visual identification can be challenging when symptoms are similar across different diseases. Delayed identification can allow diseases to spread and may negatively affect crop productivity. Farmers may also have difficulty understanding **how severely a crop is affected** and estimating how the disease may influence the expected harvest. There is a need for a system that can provide useful crop health information and help users understand possible crop losses.
C – Constraints What are the CONSTRAINTS when trying to address the situation or problem?	• The system will initially support a limited number of selected crops and diseases. • Disease prediction depends on the quality and diversity of the available crop image dataset. • Images must have sufficient quality and suitable lighting for reliable analysis. • Yield prediction depends on the availability and accuracy of agricultural and environmental data. • The system requires accurate information about crop, land, weather, soil, and other relevant factors. • AR/MR functionality depends on the capabilities of the Meta Quest device.

	• Internet connectivity may be required for AI model/API communication. • Farmer and agricultural data must be handled securely.
O – Objectives & Outcomes What are the OBJECTIVES and intended OUTCOME of this design challenge?	**Objectives** • Develop an AI-powered crop health system called **Annadatha AI**. • Detect crop diseases from uploaded or captured crop/leaf images. • Estimate the **severity of the detected disease**. • Predict expected crop yield using relevant agricultural parameters. • Estimate potential yield loss based on disease severity and other factors. • Provide general crop disease management and prevention guidance. • Integrate **Mixed Reality using Meta Quest** to visualize crop health information. • Provide information through an interactive and easy-to-understand interface. **Outcomes** • Faster identification of crop diseases. • Early understanding of disease severity. • Better estimation of expected crop yield. • Awareness of potential crop-related yield loss. • Improved understanding through Mixed Reality. • AI-assisted decision support for farmers. • An immersive learning experience for students.

P – People Who are the PEOPLE that you are trying to help? Whose problem are you trying to address? Primary Stakeholders Secondary Stakeholders	**Primary Stakeholders** • Farmers • Small-scale farmers • Agricultural workers • Students learning agriculture and crop science **Secondary Stakeholders** • Agricultural Experts • Agricultural Researchers • Agricultural Departments • Crop Consultants • Agricultural Organizations • Educational Institutions
E – Estimates What are the ESTIMATED resources required for this design challenge?	**Frontend / MR:** Unity **AR/MR Platform:** Meta Quest **MR Development:** Meta XR SDK / Unity XR tools **AI/ML:** Python, TensorFlow / PyTorch **Disease Detection:** CNN / EfficientNet / MobileNet **Severity Estimation:** Image Segmentation / Computer Vision **Yield Prediction:** Random Forest / XGBoost / Machine Learning **Backend:** Python FastAPI / Flask **Database:** MySQL / MongoDB / Firebase

	Development Team: 1 Member **Development Duration:** 8–10 Weeks
S – Scope to Explore Is there ample SCOPE to explore multiple alternatives to address the problem?	Crop Disease Detection Users can upload or capture a crop/leaf image, and the AI model identifies the crop and possible disease. Disease Severity Estimation The system estimates the percentage or level of disease affected area and categorizes the condition as: ● Mild ● Moderate ● Severe Yield Prediction The system uses factors such as: ● Crop type ● Land area ● Soil information ● Rainfall ● Temperature ● Disease ● Disease severity to estimate the expected crop yield. Yield Loss Estimation The system can provide an estimated potential reduction in crop yield based on disease severity and other agricultural factors.

	Crop Health Visualization The system provides crop disease, severity, and yield information through an interactive interface. Mixed Reality Assistance Using **Meta Quest**, the system can display AI-generated crop information as virtual panels or visual indicators positioned around the real crop. For example: **Tomato Plant** Disease: Early Blight Severity: 42% – Moderate Expected Yield: 7.2 tonnes/acre Instead of replacing agricultural experts, **ANNADATHA AI acts as an AI-assisted decision-support system**, helping farmers understand crop health and potential yield more easily.
Frame the Design Challenge	How might we design an AI-powered agricultural system that enables farmers to identify crop diseases, understand their severity, estimate potential yield and visualize crop health through Mixed Reality, while also providing students with an immersive platform to learn about crops, diseases and agricultural practices?

SIGNATURE